Liquidity Risk

Lecture 9

BANK3011

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Objectives of this week's lecture

- To explore:
  - ✓ The problem of liquidity risk, which is faced to a greater or lesser extent by all banks.
  - ✓ The methods of measuring liquidity risk, and its consequences.

- Identify:
  - ✓ How Liquidity risk is managed, including the motives for holding liquid assets
  - ✓ Specific issues associated with liability and liquidity risk management.

- To examine the developments in Liquidity Risk management in the Basel III Capital Accord

1. Liquidity risk

- ✓ Bank liquidity risk is the danger that a bank cannot meet its short-term payment obligations or fund asset growth because it cannot get cash quickly enough.
- ✓ Liquidity risk is a daily concern for Fis
- ✓ In the extreme, liquidity risk problems develop into insolvency risk problems when FIs are forced to liquidate illiquid assets at fire sale prices → Bank Failure (SVB, Credit Sussie)

1. Liquidity risk

- ✓ A bank promises to repay depositors on demand, while holding assets it cannot sell on demand.
- ✓ Therefore, a central problem in liquidity management is to assess how long the bank could keep that promise.
- ✓ FIs can usually meet the liquidity demand by:
  - ➢ Running down their cash
  - ➢ Sell off other liquid assets
  - ➢ Borrow additional funds

1. Liquidity risk

- The key issues
  - ✓ In measuring liquidity risk, it is necessary to assess what demands for funds are likely to be placed on banks because not all deposits will be demanded for repayment and not all loan and other commitments will be called upon at the same time

  - ✓ The factors in liquidity management and measurement
    - ➢ Potential cash flows
    - ➢ The Bank's capacity to borrow in the market
    - ➢ The Bank's stock of readily available high quality liquid assets

1. Liquidity risk

- The key issues
  - ✓ Thus the active management of the liquidity structure of both the asset & liabilities portfolios is required, thus involving a comprehensive response with:
    - ➢ Approach 1 – Purchased Liquidity management
    - ➢ Approach 2 – Stored Liquidity management

1. Liquidity risk

- Causes of Liquidity Risk
  - ✓ Liquidity risk is inherent in a FI's asset transformation function.
    - ➢ A DI uses a large amount of short-term liabilities to finance long-term assets
      - A bank's assets are relatively illiquid (long term loans) i.e. difficult to call in
      - A bank's liabilities are liquid i.e., can be called or withdrawn quickly

1. Liquidity risk

- Causes of Liquidity Risk
  - ✓ Liability-side liquidity risk:
    - ➢ The FI may not have enough cash to meet the requests for withdrawals.
  - ✓ Asset-side liquidity risk:
    - ➢ The FI may not have enough cash to fund the exercise of loan commitment or other commitments for lending (e.g. Covid-19 dash for cash)
    - ➢ The value of a FI's investment portfolio may fall unexpectedly (e.g. SVB)

1. Liquidity risk

- Example of the balance sheet of commercial banks

Assets			Liabilities*		
Total cash assets	$ 397.6	4.14%	Total deposits	$6,383.0	73.96%
Total securities	2,731.1	28.44	Borrowings	1,954.3	22.64
Total loans	5,589.3	58.21	Other liabilities	293.2	3.40
Other assets	884.3	9.21	Total liabilities	$8,630.5	
Total assets	$9,602.3				

2. Managing liquidity risk

- Liability-Side Liquidity Risk
  - ✓ A DI knows that only a small proportion of its deposits will be withdrawn on a given day and part of these withdrawals are also offset by the inflow of new deposits. The difference between deposit withdrawals and deposit additions is called the **net deposit drain**.
  - ✓ Over time, a DI manager can normally predict the probability distribution of net deposit drain with a good degree of accuracy.
  - ✓ **Most demand deposits** act as consumer **core deposits**, providing a relatively **stable** or **long-term** source of funds for the DI.

2. Managing liquidity risk

- Managing Deposit Drains using Purchased Liquidity Management
  - ✓ To manage, a DI can purchase liquidity by
    - ➢ Borrowing on the market for purchased funds to cover cash outflows
      - ▪ The Federal Funds Market (interbank cash market) or Repurchase Agreement (Repo) Market
    - ➢ Issuing wholesale CDs or even sell some notes and bonds.
  - ✓ The benefit is that it insulates the size and composition of the asset side of the balance sheet from normal deposit drains.

2. Managing liquidity risk

- Managing Deposit Drains using Purchased Liquidity Management
  - ✓ The cost is that
    - ➢ it can be expensive for the DI since it has to pay the usually higher market rates for funds in the wholesale money market to offset net drains on low-interest-bearing deposits.
    - ➢ Furthermore, the availability of these funds can be limited when the lenders are concerned about the solvency of the DI and in times of financial crises

2. Managing liquidity risk

- Managing Deposit Drains using Purchased Liquidity Management
  - ✓ Example: the DI purchased (borrowed) $5 million to meet the deposit drain.

Panel A: Balance Sheet Immediately before and after Deposit Drain

Before the Drain

Assets		Liabilities	
Assets	100	Deposits	70
		Borrowed funds	10
		Other liabilities	20
	100		100

After the Drain

Assets		Liabilities	
Assets	100	Deposits	65
		Borrowed funds	10
		Other liabilities	20
	100		95

Panel B: Adjusting to a Deposit Drain through Purchased Liquidity Management

Assets		Liabilities	
Assets	100	Deposits	65
		Borrowed funds	15
		Other liabilities	20
	100		100

2. Managing liquidity risk

- Managing Deposit Drains using Stored Liquidity Management
  - ✓ A DI can also utilize its stored liquidity to meet positive net deposit drains.
    - ➢ Run down cash assets.
    - ➢ Sell assets.
  - ✓ The benefit is that it does not rely on the availability of funds on the market.
  - ✓ The cost to the DI is that, apart from decreased asset size, it must hold excessively low-return assets on the balance sheet and thus forgo the returns that it could otherwise earn by investing these funds in loans and other higher-income-earning assets.

Therefore banks typically combine purchased & stored liquidity management approaches

2. Managing liquidity risk

- Managing Deposit Drains using Stored Liquidity Management
  - ✓ Example: The DI run down $5 million cash to meet the withdraw demand. The balance sheet shrinks by $5 million.

Panel A: Balance Sheet Immediately before Deposit Drain

Assets		Liabilities	
Cash	9	Deposits	70
Other assets	91	Borrowed funds	10
		Other liabilities	20
	100		100

Panel B: Adjusting to a Deposit Drain through Stored Liquidity Management

Assets		Liabilities	
Cash	4	Deposits	65
Other assets	91	Borrowed funds	10
		Other liabilities	20
	95		95

2. Managing liquidity risk

- Managing Asset-Side Liquidity Risk
  - ✓ Can use both purchased and stored liquidity management methods.
  - ✓ Example: An exercise of $5 million loan commitment.

Panel A: Balance Sheet Immediately before and after Exercise

(a) Before Exercise

Cash	9	Deposits	70
Other assets	91	Borrowed funds	10
		Other liabilities	20
	100		100

(b) After Exercise

Cash	9	Deposits	70
Other assets	96	Borrowed funds	10
		Other liabilities	20
	105		100

Panel B: Adjusting the Balance Sheet to a Loan Commitment Exercise

(a) Purchased Liquidity Management

Cash	9	Deposits	70
Other assets	96	Borrowed funds	15
		Other liabilities	20
	105		105

(b) Stored Liquidity Management

Cash	4	Deposits	70
Other assets	96	Borrowed funds	10
		Other liabilities	20
	100		100

3. Measuring liquidity risk

- 1. Net Liquidity Statement
  - ✓ It is important that a DI manager can measure its liquidity position on a daily basis. A useful tool today is a net liquidity statement which lists the sources and uses of liquidity and thus provides a measure of the DI's liquidity position and helps to identify when liquidity issues may arise

3. Measuring liquidity risk

- 1. Net Liquidity Statement
  - ✓ The DI can obtain liquid funds in three ways.
    - ➢ First, It can sell its liquid assets with little price risk and low transaction cost.
    - ➢ Second, it can borrow funds in the money/purchased funds market up to a maximum amount. The market would impose such a limit based on the DI's debt capacity.
    - ➢ Third, it can use any excess cash reserve over and above the amount held to meet regulatory imposed reserve requirements.

3. Measuring liquidity risk

- 1. Net Liquidity Statement
  - ✓ The net liquidity statement is designed to measure the sources of and current utilization of liquidity
  - ✓ Sources of Liquidity (recall from previous slide):
    - ➢ Cash type assets
    - ➢ Maximum amount of borrowed funds available (estimate)
    - ➢ Excess cash reserves
  - ✓ Current Liquidity Utilization:
    - ➢ Borrowed or money market funds already utilized
    - ➢ Assets already subject to Repo

3. Measuring liquidity risk

- 1. Net Liquidity Statement

✓ Example: Net Liquidity Statement

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TABLE 12–6
Net Liquidity Position (in millions of dollars)

Sources of Liquidity	
1. Total cash-type assets	$ 2,000
2. Maximum borrowed funds limit	12,000
3. Excess cash reserves	500
Total	$14,500
Uses of Liquidity	
1. Funds borrowed	$ 6,000
2. Federal Reserve borrowing	1,000
Total	7,000
Total net liquidity	$ 7,500

- 2. Peer Group Ratio Comparisons
  - ✓ Compare certain key ratios and balance sheet feature of the DI with those of DIs of a similar size and geographic location.
    - ➢ Loan to deposit ratio
    - ➢ Borrowed funds to total assets
    - ➢ Commitment to lend to total assets.
  - ✓ A high ratio of loans to deposits and borrowed funds to total assets means that the DI relies heavily on the short-term money market rather than core deposits to fund loans.

- 3. Liquidity Index
  - ✓ This index measures the potential losses a FI could suffer from a sudden or fire-sale disposal of assets compared with the amount it would receive at a fair market value established under normal market (sale) conditions – which might take a lengthy period of time as a result of a careful search and bidding process.

$$I = \sum_{i=1}^{N} w_i \left(\frac{P_i}{P_i*}\right)$$

where w_i is the weight of asset i in total asset portfolio, P_i is asset i's fire-sale price and P_i* is asset i's fair market price.

  - ✓ $0<I\leq 1$ and the higher the I the more liquid the DI's portfolio of assets.

- 3. Liquidity Index
  - ✓ Example: Calculation of Liquidity Index
  - ✓ Suppose a DI has two assets: 50 percent in one-month T-bills and 50 percent in real estate loans. If DI must liquidate its T-bills today, it receives $99 per $100 of face value. If it can wait to liquidate them on maturity (in one month's time), it will receive $100 per $100 face value. If the DI has to liquidate its real estate loans today, it receives $85 per $100 face value. Liquidation at the end of the month (closer to maturity) will produce $92 per $100 face value.

- 3. Liquidity Index
  - ✓ Example: Calculation of Liquidity Index
  - ✓ The one-month liquidity index for this DI's asset portfolio is:

$$I = \left(\frac{1}{2}\right)\frac{.99}{1.00} + \left(\frac{1}{2}\right)\frac{.85}{.92} = 0.957$$

  - ✓ Suppose, alternatively, that a slow or thin real estate market caused the DI to be able to liquidate the real estate at only \$65 per \$100 face value. The one-month liquidity index for the DI's asset portfolio is:

$$I = \left(\frac{1}{2}\right)\frac{.99}{1.00} + \left(\frac{1}{2}\right)\frac{.65}{.92} = 0.848$$

- 4. Financing Gap and the Financing Requirement
  - ✓ Although demand deposits can be withdrawn anytime, most depositors do not do so in normal conditions. As a result, most demand deposits stay at DIs for quite long periods – often two years or more. Thus, they are considered to be a core source of funding.
    - ➢ $Financing\ Gap = Average\ loans - Average\ deposits$

- 4. Financing Gap and the Financing Requirement
  - ✓ A positive financing gap must be funded by either running down cash and liquid assets or borrowing on the market
    - ➢ $Financing\ Gap = -Liquid\ assets + Borrowed\ funds$
    - ➢ $Financing\ gap + Liquid\ assets = Financing\ requirement$
  - ✓ The larger a DI's financing gap and liquid asset holdings, the larger the amount of funds it needs to borrow in the money markets and the greater is the exposure to liquidity problems from such an reliance.

3. Measuring liquidity risk

- 4. Financing Gap and the Financing Requirement

✓ Example

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TABLE 12–8
Financing Requirement of a DI (in millions of dollars)

Assets		Liabilities	
Loans	$25	Core deposits	$20
Liquid assets	5	Financing requirement (borrowed funds)	10
Total	$30	Total	$30
		Financing gap	5

✓ This balance sheet illustrate the following relation

➢ $Financing\ \ gap$ ($5 $million$)+$Liquid\ \ assets$ ($5 $million$)=$Financing\ Requirement$ ($10 $million$)

- 4. Financing Gap and the Financing Requirement
  - ✓ Example
    - ➢ A widening financing gap can warn of future liquidity problems for a DI since it may indicate increased deposit withdrawals and increasing loans due to increased exercise of loan commitment.
    - ➢ If the DI does not reduce its liquid assets, the manager must resort to more money market borrowings.

- 4. Financing Gap and the Financing Requirement

✓ As these borrowings rise, sophisticated lenders in the money market may be concerned about the DI's creditworthiness. They may react by imposing higher risk premiums on borrowed funds or establishing stricter credit limits by not rolling over funds lent to the DI. If the DI's financing requirements exceed such limits, it may become insolvent.

✓ The financing gap can be used in peer group comparisons or trends within a bank.

4. Liquidity planning and management

- Liquidity Planning and Management
  - ✓ Identify which types of depositors are likely to withdraw across industry sectors and countries if borrowing offshore.
    - ➢ The composition of the depositor base will affect the severity of funding shortfalls.
  - ✓ Identify the size of deposits and withdrawls over various time horizons (weeks, months, and years)

5. Bank runs

- Bank Runs are costly
  - ✓ Bank run is a sudden and unexpected increase in deposit withdrawals from a DI
  - ✓ Can arise due to:
    - ➢ Concern about a individual bank's solvency.
    - ➢ The failure of a related financial institution.
    - ➢ Sudden changes in investor preferences towards non-bank investments

5. Bank runs

- Bank Runs are costly
  - ✓ Can arise due to:
    - ➢ Demand deposits are first come first served. Depositors' place in line to receive payment in the event of a bank failure is important.
    - ➢ Bank panic: systemic or contagious bank run, justified or not are disruptive and can force DIs into insolvency
      - ▪ As seen in the GFC, Northern Rock and IndyMac and WaMu
      - ▪ More recently in Greece and Cyprus

6. Basel III – liquidity standard

- The Key Initiatives Basel III – Summary of New Liquidity Standard
  - ✓ A centre-piece of the post-GFC reforms reflected in Basel III is a new focus on reducing liquidity risk with two key ratios:

  - ✓ 1. Liquidity Coverage Ratio (LCR)
    - ➢ Requires high quality liquid assets to be held that are able to be sold to cover cash-outflows under a 30 day stress scenario.
    - ➢ Committed liquidity facility from RBA in Australia to help banks meet the new liquidity standards
    - ➢ Fully in place by Jan 2015

6. Basel III – liquidity standard

- The Key Initiatives Basel III – Summary of New Liquidity Standard
  - ✓ 2. Net Stable Funding Ratio (NSFR)
    - ➢ Aims to ensure that long term assets are funded by at lest a minimum of stable liabilities (matches uses and sources of funds)
    - ➢ Fully in place by Jan 2018
  - ✓ Both ratios need to be > 100% going forwards
    - ➢ Banks are not exactly happy with the additional regulatory burden!

6. Basel III – liquidity standard

- The Liquidity Coverage Ratio Under Basel III
  - ✓ The liquidity coverage ratio (LCR) aims to ensure that a DI maintains its high-quality assets that can be converted into cash to meet liquidity needs for a 30-day time horizon under an “acute liquidity stress scenario” specified by supervisors.

$$LCR = \frac{Stock\ of\ high\ quality\ liquid\ assets}{Total\ net\ cash\ outflows\ over\ the\ next\ 30\ calendar\ days} \geq 100\%$$

6. Basel III – liquidity standard

- The Liquidity Coverage Ratio Under Basel III
  - ✓ The stock of high quality liquid assets is defined as follows
    - ➢ Low risk, ease of valuation, low correlation with risky assets, listed on developed exchange
    - ➢ Assets traded in active market with low volatility

6. Basel III – liquidity standard

- Net Stable Funding Ratio Under Basel III
  - ✓ The net stable funding ratio (NSFR) requires a minimum amount of stable funding be held over a one-year time horizon based on the liquidity risk factors assigned to liquidity exposures of on- and off-balance sheet assets.

$$NSFR = \frac{Available\ amount\ of\ stable\ funding}{Required\ amount\ of\ stable\ funding} > 100\%$$

6. Basel III – liquidity standard

- Net Stable Funding Ratio Under Basel III
  - ✓ Available stable funding includes
    - ➢ Bank capital
    - ➢ Preferred stock with a maturity > 1 year
    - ➢ Liabilities with maturities > 1 year
    - ➢ The portion of retail deposits and wholesale deposit expected to stay with bank during a period of idiosyncratic stress.

Summary

- Liquidity risk is a key concern for FI managers and is an inherent risk in financial intermediation
- Can be measured with different metrics – net liquidity statement; peer group ratios; liquidity index; Financing gap and requirement
- FIs can use purchased or stored liquidity management approaches to deal with liquidity risk but also need to diligently manage tradeoffs in risk/return
- Recent banking regulatory reforms have focused on new liquidity metrics (LCR and NSFR) to help reduce liquidity risk in FIs, already partially in place around the world
- We will see how deposit insurance schemes help next week…